

- El. polje v snovi
- v snovi vezan naboj, vpliv zun. nabojev
 - vezan naboj
- $$q_v(\vec{r}, t) = \sum_i e_i \delta^3(\vec{r} - \vec{r}_i)$$
- $$\rightarrow \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} + \frac{\rho_v}{\epsilon_0}$$
- Polarizacija
- $$\rho_v = -\nabla \cdot \vec{P} \rightarrow \nabla \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho$$
- Gostota el. polja
- $$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \rightarrow \nabla \cdot \vec{D} = \rho$$
- Maxwellove en. za snov
- Konstitutivna relacija za el. polje v snovi
- $$\vec{P} = \vec{P}(\vec{D})$$
- poljubna konst. rel.
- najbolj poljub. izotropni lin. snovi:
- $$\vec{P} = \chi_e \epsilon_0 \vec{E} + o(D^2)$$
- $$\chi_e = 1 - \frac{1}{\epsilon_r} \rightarrow$$
- diel. konst.
- $$\vec{D} = \epsilon_0 \vec{E} + (1 - \frac{1}{\epsilon_r}) \vec{D} \rightarrow \vec{D} = \epsilon \vec{E}$$

- Klasifikacija snovi glede na odziv na el. polje
- dielektriki (končna vrednost ϵ)
 - lahko skladiščijo energijo
 - neidealni dielektriki: imajo $\epsilon = \epsilon(\omega)$
 - prevodniki ($\epsilon \rightarrow \infty$)
 - idealno senčijo el. polje

- Magnetno polje v snovi
- v snovi se ustvarijo vezani tokovi (kv.moh. izvor)
 - vezan tok
- V snovi imamo:
- gostoto zunanjegega toka $\vec{j}$
 - gostoto vezanega toka $\vec{j}_v = \sum_i \vec{j}_i \delta^3(\vec{r} - \vec{r}_i)$
- $$\rightarrow \nabla \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

- Magnetizacija
- $$\vec{j}_v = \nabla \times \vec{M} + \frac{\partial \vec{P}}{\partial t}$$
- $$\nabla \cdot \vec{j}_v + \frac{\partial \rho_v}{\partial t} = \nabla \cdot (\nabla \times \vec{M} + \frac{\partial \vec{P}}{\partial t}) + \frac{\partial}{\partial t} (\nabla \cdot \vec{P}) = 0$$
- v enodimenz. konst. en.
- $$\nabla \times (\frac{\vec{B}}{\mu_0} - \vec{M}) = \vec{j} + \frac{\partial \vec{P}}{\partial t} \rightarrow \vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}$$
- jakost magn. polja
- $$\rightarrow$$
- Max. en. v snovi:
- $\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$
- Konstitutivna relacija za mag. polje v snovi
- $$\vec{M} = \vec{M}(\vec{H})$$
- poljubna konst. relacija
- v izotropni, lin. snovi in za šibka polja: $\vec{M} = \chi_m \vec{H} + o(H^2)$ mag. susceptibilnost
 - $\chi_m = \mu - 1$ mag. permeabilnost
- $$\rightarrow \vec{H} = \frac{\vec{B}}{\mu_0} - \chi_m \vec{H} \rightarrow \vec{B} = \mu \vec{H}$$
- $$\vec{H} = (1 - \frac{1}{\mu}) \frac{\vec{B}}{\mu_0}$$

- Magnetizacija in gostota mag. dip. momenta
- $\vec{M}$ je volumska gostota mag. dip. momenta (brez izpolnjevanja)

- Klasifikacija snovi glede na odziv na mag. polje
- $\rightarrow$ feromagnetiki
- imajo permanentno magnetizacijo, ki je neodv. od zun. mag. polja
 - $\vec{M} = \vec{M}(T)$
- $\rightarrow$ diamagnetiki
- $\vec{M}$ samo ob prisotnosti zun. mag. polja
 - $\vec{M}$ kaže v nasprotno smer kot mag. polje ($\chi_m < 0$)
- $\rightarrow$ paramagnetiki
- $\vec{M}$ samo ob prisotnosti polja
 - $\vec{M}$ kaže v isto smer kot mag. polje (ga ojačuje)

- Maxwellove en. v snovi
- | | |
|--|---|
| $\nabla \cdot \vec{D} = \rho$ | $\nabla \cdot \vec{B} = 0$ |
| $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ | $\nabla \times \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$ |
- +2 konst. relaciji:
- $$\vec{P} = \vec{P}(\vec{D}) \rightarrow \vec{D} = \epsilon \vec{E}$$
- $$\vec{M} = \vec{M}(\vec{H}) \rightarrow \vec{B} = \mu \vec{H}$$
- v izotropni lin. snovi
- v anizotropni snovi: $\epsilon \rightarrow \underline{\epsilon} \rightarrow \vec{D} = \underline{\epsilon}_0 \underline{\epsilon} \vec{E}$
- tenzor
- $$\underline{\mu} \rightarrow \underline{\mu} \rightarrow \vec{B} = \underline{\mu} \vec{H}$$
- v nelinearnih snovih: $\vec{D} = \underline{\epsilon}_0 \vec{E} + \underline{\epsilon}_2 \vec{E} \vec{E} + \dots$
- $$\vec{B} = \underline{\mu}_0 \vec{H} + \underline{\mu}_2 \vec{H} \vec{H} + \dots$$
- tenzor 2. reda

Ohranitveni zakoni v snovi

- Ohranitev energije
- konst. en.: $\frac{\partial W}{\partial t} + \nabla \cdot (\vec{P} + \vec{j} \cdot \vec{E}) = 0$; $W = \int_V \vec{E}(\vec{r}) \cdot d\vec{D} + \int_V \vec{H}(\vec{r}) \cdot d\vec{B}$ gostota energije
- $$\vec{P} = \epsilon \vec{E} \cdot \vec{H}$$
- Poyntingov vektor
- Energija EMP v snovi
- električno polje
- $$W - W_0 = \int_V w dV - \int_V w_0 dV = \int_V \int_0^{\vec{D}} \vec{E} d\vec{D} dV - \int_V \int_0^{\vec{D}_0} \vec{E}_0 d\vec{D}_0 dV = \int_V \int_0^{\vec{D}} (\vec{E} - \vec{E}_0) d\vec{D} dV = \int_V \int_0^{\vec{D}} (\vec{E} - \vec{E}_0) d\vec{D} dV$$
- $$W - W_0 = \frac{1}{2} \int_V (\vec{E} - \vec{E}_0) \cdot \vec{D} dV + \frac{1}{2} \int_V (\vec{E} + \vec{E}_0) \cdot (\vec{D} - \vec{D}_0) dV = \frac{1}{2} \int_V (\vec{E} - \vec{E}_0) \cdot \vec{D} dV - \frac{1}{2} \int_V (\vec{E} - \vec{E}_0) \cdot \vec{D}_0 dV + \frac{1}{2} \int_V \vec{E}_0 \cdot \vec{D}_0 dV$$
- $\vec{D} - \vec{D}_0 = 0$ (zun. izvori se ne spreminjajo)
- prevedemo na površ. integral, kjer stena od snovi $\vec{D} = \vec{D}_0 \rightarrow$ integral = 0
- $$\rightarrow W - W_0 = \frac{1}{2} \int_V \epsilon (\vec{E} - \vec{E}_0) \cdot \vec{E} dV = -\frac{1}{2} \int_V \vec{P} \cdot \vec{E}_0 dV$$
- v snovi
- BREZ snovi
- magnetno polje
- $$W - W_0 = \frac{1}{2} \int_V \vec{H} \cdot \vec{B} dV - \frac{1}{2} \int_V \vec{H}_0 \cdot \vec{B}_0 dV = \frac{1}{2} \int_V (\vec{B} - \vec{B}_0) \cdot \vec{H} dV + \frac{1}{2} \int_V (\vec{B}_0 + \vec{B}) \cdot \vec{H} dV = \dots$$
- podobno kot prej
- $$W - W_0 = \frac{1}{2} \int_V \mu (\vec{H} - \vec{H}_0) \cdot \vec{H} dV = \frac{1}{2} \int_V \vec{H} \cdot \vec{B}_0 dV$$

- Cauchyjae kont. et. za GK!

$$\frac{\partial g_i}{\partial t} - \frac{\partial T_A}{\partial x_i} + f_i = 0$$

$$g = \vec{D} \times \vec{B} \text{ gestohlen GR}$$

$$T_k = E_k D_k - \frac{1}{2}(\vec{E} \cdot \vec{D})\delta_{ik} + B_k H_k - \frac{1}{2}(\vec{B} \cdot \vec{H})\delta_{ik} \text{ napredstvi dave, } E \text{ i } B \text{ polja}$$

- Robni pogoji za Maxwell. valacije

$$\begin{array}{|l|l|l|} \hline D_{2N} - D_{2N}^* = \delta & B_{2N} - B_{2N}^* = 0 & \delta \text{ površinska gost. nabija} \\ E_{42} - E_{42}^* = 0 & H_{42} - H_{42}^* = K & K \text{ površinska gost. toka} \\ \hline \end{array}$$

$$\vec{B}: \quad \vec{B} = 0 \Rightarrow \int_V \vec{B} \cdot d\vec{s} = 0 = \oint_{\partial V} \vec{B} \cdot d\vec{s} = \oint_{\partial V} (B_1 \hat{e}_1 + B_2 \hat{e}_2 + B_3 \hat{e}_3) \cdot d\vec{s} = \oint_{\partial V} B_1 \hat{e}_1 \cdot d\vec{s} + \oint_{\partial V} B_2 \hat{e}_2 \cdot d\vec{s} + \oint_{\partial V} B_3 \hat{e}_3 \cdot d\vec{s} = 0 \Rightarrow B_1 \hat{e}_1 \cdot d\vec{s} + B_2 \hat{e}_2 \cdot d\vec{s} + B_3 \hat{e}_3 \cdot d\vec{s} = 0$$

$$\vec{D} \cdot \vec{B} = g \Rightarrow \int_V \vec{D} \cdot d\vec{A} = \int_V g dV = \int dA \Delta z \Rightarrow \vec{D}_1 \cdot \hat{a}_1 + \vec{D}_2 \cdot \hat{a}_2 = (\vec{D}_1 \cdot \vec{D}_2) \hat{a}_1 = d$$

$$\vec{E} \cdot \frac{\nabla \vec{E}}{\nabla E} \quad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \rightarrow \int \nabla \times \vec{E} \cdot d\vec{S} = -\int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} = -\int \frac{\partial}{\partial t} (\vec{B} \cdot d\vec{S}) \Rightarrow \vec{E} \cdot \vec{E} + \vec{E} \cdot \vec{E} = (\vec{E} \cdot \vec{E}) \cdot \vec{E} = 0$$

$$\vec{H}_1 \cdot \vec{E}_1 + \vec{H}_2 \cdot \vec{E}_2 = (\vec{H}_1 - \vec{H}_2) \cdot \vec{E}_1 = K$$